



Bangladesh University of
Business and Technology (BUBT)

PROJECT PROPOSAL

On

[Project Name: PowerPuff Quest]

[Submission Date: June 18, 2026]

Submitted By:

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Submitted To

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1. Needs/Problems

New students in C programming lack exciting projects to learn concepts such as file handling, decision making with arrays creating functions and string manipulation. This makes them lose motivation. Usual projects like making a calculator or text editor are not engaging, resulting in learning and easy forgetting.

The problems this project will solve are:

- There are not fun and interactive ways to learn C programming that cover important topics like login systems, loops, decision making, file handling and data management in one application.
- Students struggle to connect programming concepts, such as scanf() fscanf() switch-case, arrays and functions to real-world software development.
- There are no C console games, like role-playing games that can be played on any computer without extra libraries or frameworks.
- Most teaching projects focus on one concept at a time rather than combining multiple concepts to create something practical and fun.
- Students need a platform to practice programming concepts while playing a game with features, like character selection, combat, progression and account management.

2. Goals/Objectives

PowerPuff Quest is a game where you get to play roles. It was made using the C programming language. The main idea of **PowerPuff Quest** is to teach people about the basics of C programming in a way. The system lets people create an account and log in. It stores usernames and passwords in a file called users.txt. You can choose which **PowerPuff Girl** you want to be. You can be Blossom, you can be Bubbles. You can be Buttercup. Each PowerPuff girl is unique. **PowerPuff Girls** have their strengths and weaknesses. In **PowerPuff Quest** you fight against monsters in five levels. The monsters get harder to beat as you go along. You can attack the monsters. You can also heal yourself to stay alive. The game uses colours to make it more fun to look at. This is done using the computer's abilities so it does not need any help from other programs. PowerPuff Quest is a game that uses colours to make it fun.

3. Existing Project Evaluation

1. Tiny C RPG (GitHub, 2021)

A text-based dungeon crawler developed in C.

Shortcomings: No user authentication, file persistence, or colour output. The project uses a single-file structure with hardcoded player statistics, making future expansion difficult.

2. BattleArena-C (SourceForge, 2019)

A two-player turn-based battle game.

Shortcomings: Requires two human players, lacks monster AI progression and character selection, and is limited to Linux systems.

3. SimpleLogin-C (CodeProject, 2020)

A file-based login system written in C.

Shortcomings: Provides only authentication functionality without any game features. Passwords are stored in plain text with no basic security or obfuscation.

4. Features / Scopes

The game has a system where users can register and log in. Their information is saved in a file called users.txt.

The game has three characters to choose from. Each character has abilities to start with. The characters are different. They have their own strengths. The game has five levels of monsters. Each level gets harder than the one. The monsters get tougher as you go through the levels. The levels are five. They keep getting harder. When you fight you can choose to Attack or Heal. You get health when you finish each level. This makes it easier to play. The game uses colours to make it look nicer. The colours make the game more fun to play. The game ends with a game over screen or a victory screen. There is also a menu where you can register, login or play. You can choose what you want to do from the menu. The menu has options to register, login and play.

5. Development Model

The 'Iterative and Incremental Model' was selected because it allows the project to be developed step by step. First, the login system was created, then character selection, followed by the battle system, and finally the user interface. Each part was tested before moving to the next one. This model is suitable for a small academic project because it makes development easier and helps find and fix problems early.

6. Feasibility Analysis & Possible Risks

Technical Feasibility

The project is developed using the C programming language and standard libraries such as **stdio.h**, **string.h**, and **stdlib.h**. These libraries are available in all standard C compilers, so no extra software is needed.

Operational Feasibility

The game is easy to use because it has a simple menu-driven interface. Users only need basic computer knowledge to play the game, and no database setup is required since data is stored in text files.

Economic Feasibility

The project has no additional cost. Free tools such as **GCC**, **Code::Blocks**, or **VS Code** are used, so no paid software or hardware is required.

Possible Risks

Some older Windows terminals may not display colours correctly. Also, passwords are stored as plain text, which is less secure. If the project becomes larger in the future, the code may be harder to manage. These issues can be improved in future versions.

7. Requirements

- **Hardware Requirements:** A computer or laptop with basic specifications is enough to run the project.
- **Software Requirements:** A C compiler and any code editor, such as Code::Blocks or VS Code, are required.
- **Programming Language:** The project is developed using the C programming language.
- **Tools:** Code::Blocks or VS Code is used for coding, and users.txt is used to save user login information.
- **Libraries Used:** The project uses stdio.h, string.h and stdlib.h header files.

8. Activity and Time Schedule / Project Breakdown with Time Estimation

#	Stages/ Tasks	Efforts, man-hours
Stage 1	Analysis & Design	
1.1	Requirements analysis and game design	3
1.2	Flowchart and planning	2
Stage 2	Implementation	
2.1	Registration and login module	4
2.2	Character selection	1
2.3	Battle system	5
2.4	Main menu and colour interface	3
2.5	Monster level system (5 levels with increasing difficulty)	3
2.6	Healing mechanism and game balance logic	3
2.7	Win/Lose conditions and game flow control	2
Stage 3	Testing & QA	
3.1	Testing and bug fixing	6
Stage 4	Documentation	
4.1	Final documentation and submission	4
	Total estimated efforts: 38 man-hours	

9. Budget

As this is an academic project, no additional cost is required. Free software and tools are used during development.

Project Item	Amount	Unit Price
GCC Compiler	1	0
Code::Blocks / VS	1	0
GitHub	1	0
Development Cost	1	0
Grand Total	Total=	BDT 0

10. Development Team & Stakeholders

Id	Name	Designation	Responsibilities
20254103260	Rubaba Binte Mahbub	Team Leader/ QA & UI Developer	Team managing, co-ordinating among members, registration & login system, file handling, Combat system testing, color UI design (ANSI interface), debugging support, documentation assistance, and feature improvement suggestions.
20254103252	Rezowana Hossain	Co-leader/ System Designer	Project planning, game design, main menu development, game flow control, file handling, and full system integration.
20254103266	Mehreen Bintay Yousuf Simy	Back-end Programmer / Gameplay Logic Developer	Battle engine implementation (turn-based system), file handling, monster AI logic, HP/attack balancing, level progression system, and core gameplay mechanics

11. Limitations & Future Works

Current Limitations

The system currently stores passwords in plain text without any encryption, which is not secure. The game does not support save or load functionality, so all progress is lost after exiting. Input validation is limited, and invalid inputs may cause unexpected behavior. Additionally, ANSI color support may not work properly on all Windows terminals without extra configuration.

Planned Future Enhancements

In future updates, password hashing (e.g., SHA-256) will be implemented to improve security. A save/load system will be added to preserve player progress between sessions. The game will also include special abilities for each character to enhance gameplay. A multiplayer mode using socket programming is planned for extended functionality. Finally, the project may be upgraded to a graphical interface using SDL2 or a web-based system using CGI/C.

12. References

- [1] B. W. Kernighan and D. M. Ritchie, *The C Programming Language*, 2nd ed. Prentice Hall, 1988.
- [2] GNU Project, "The GNU C Library Reference Manual," gnu.org. [Online]. Available: <https://www.gnu.org/software/libc/manual/>
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- [4] Cartoon Network, "The Powerpuff Girls," Hanna-Barbera Productions, 1998–2005.